SMOOTH MORPHISMS OF LOCALLY NOETHERIAN SCHEMES PRESERVE MULTIPLICITIES OF COMPONENTS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file stacks-geometry.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Span recomputation by label pairing (LANE_BRIEF 10.2): in this chapter the author's proof environment follows the statement environment as a standalone block; the spec covers the complete environment pair (statement span, proof span) as recorded in the pool candidate.

Editorial note. Occupation scan: combined claim_key + statement_anchor + generic-line scan over all registration files (active and withdrawn) found 0 hits for stacks-geometry.tex (ge-occupied.json, 2026-10-14); 207 route=geometry / chapter-list field hits in focused-seeds are alias noise (route name and alphabetical chapter enumerations), excluded per the ladder alias-blind-spot caveat.

Source reference. Original reference 02M1: sources/stacks-chow/tags.txt, source lines 3428–3428. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 056T: sources/stacks-chow/tags.txt, source lines 6721–6721. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let $U \to X$ be a smooth morphism of locally Noetherian schemes. Let T' is an irreducible component of U. Let T be the irreducible component of X obtained as the closure of the image of T'. Then $m_{T',U} = m_{T,X}$.*

Proof. Write ξ' for the generic point of T', and ξ for the generic point of T. Let $A = \mathcal{O}_{X,\xi}$ and $B = \mathcal{O}_{U,\xi'}$. We need to show that $\text{length}_A A = \text{length}_B B$. Since $A \to B$ is a flat local homomorphism of rings (since smooth morphisms are flat), we have

$$\text{length}_A(A)\text{length}_B(B/\mathfrak{m}_A B) = \text{length}_B(B)$$

by Algebra, Lemma 02M1. Thus it suffices to show $\mathfrak{m}_A B = \mathfrak{m}_B$, or equivalently, that $B/\mathfrak{m}_A B$ is reduced. Since $U \to X$ is smooth, so is its base change $U_\xi \to \operatorname{Spec} \kappa(\xi)$. As U_ξ is a smooth scheme over a field, it is reduced, and thus so its local ring at any point (Varieties, Lemma 056T). In particular,

$$B/\mathfrak{m}_A B = \mathcal{O}_{U,\xi'}/\mathfrak{m}_{X,\xi}\mathcal{O}_{U,\xi'} = \mathcal{O}_{U_\xi,\xi'}$$

is reduced, as required. □

Date: stacks-project:stacks-geometry.tex:local-snapshot-20261012; GNU FDL 1.2 or later.